13.6.1 **Summary** 22/6/2026

Given a level surface $S : F(x\,y\,z) = c$

1) a) Tangent Plane (to the surf S at point P)

- thro point P
- $\perp$ $\underset{\sim}{N} = \underset{\sim}{\nabla} F(P)$

b) Normal Line (to the surf S at point P)

- thro point P
- $\parallel \underset{\sim}{d} = \underset{\sim}{\nabla} F(P)$

Tangent Plane to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0 \qquad (1)$$

Normal Line to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$x = x_0 + f_x(P_0)t, \qquad y = y_0 + f_y(P_0)t, \qquad z = z_0 + f_z(P_0)t \qquad (2)$$

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2) $D_{\underset{\sim}{u}} f$ = rate of change of f in the direction of $\underset{\sim}{u}$ per unit change in S $\left(\equiv \frac{df}{ds}\right)$ [Sect 3.5]

(#) Small change in f $\equiv$ (rate of change of f) $\cdot$ (small change in S)

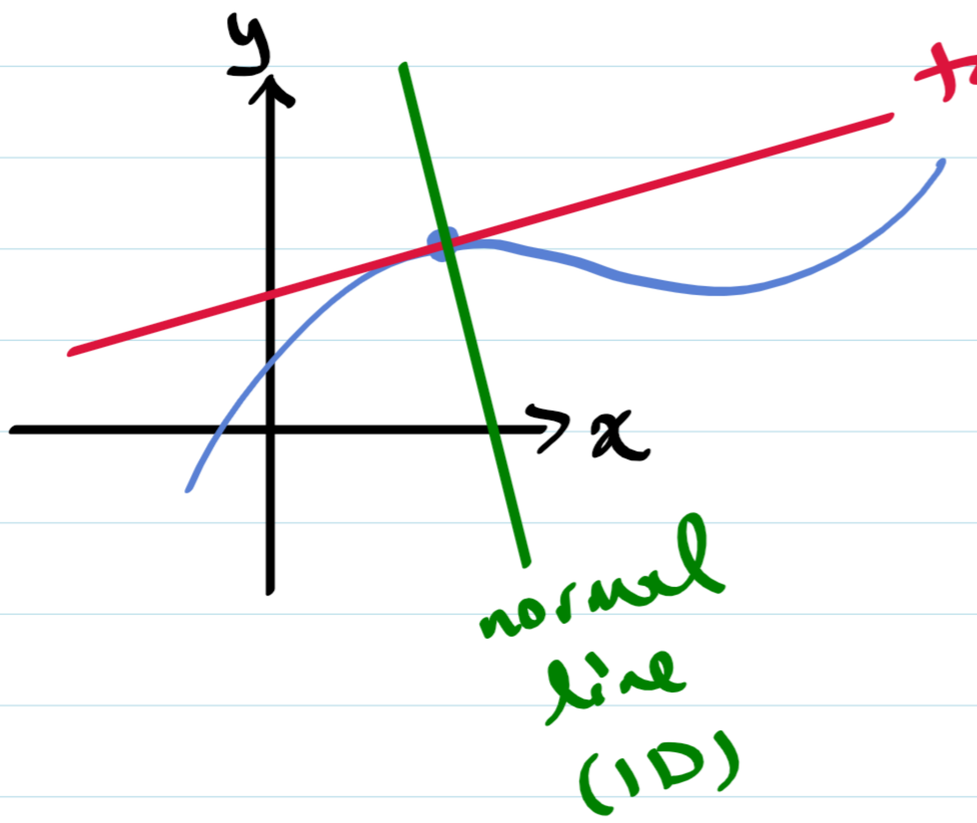
(ES) In Calculus I $\quad \Delta f \equiv \frac{df}{ds} \Delta S$

Δf: small change in f; ΔS: small change in S

Decomposition of the Space

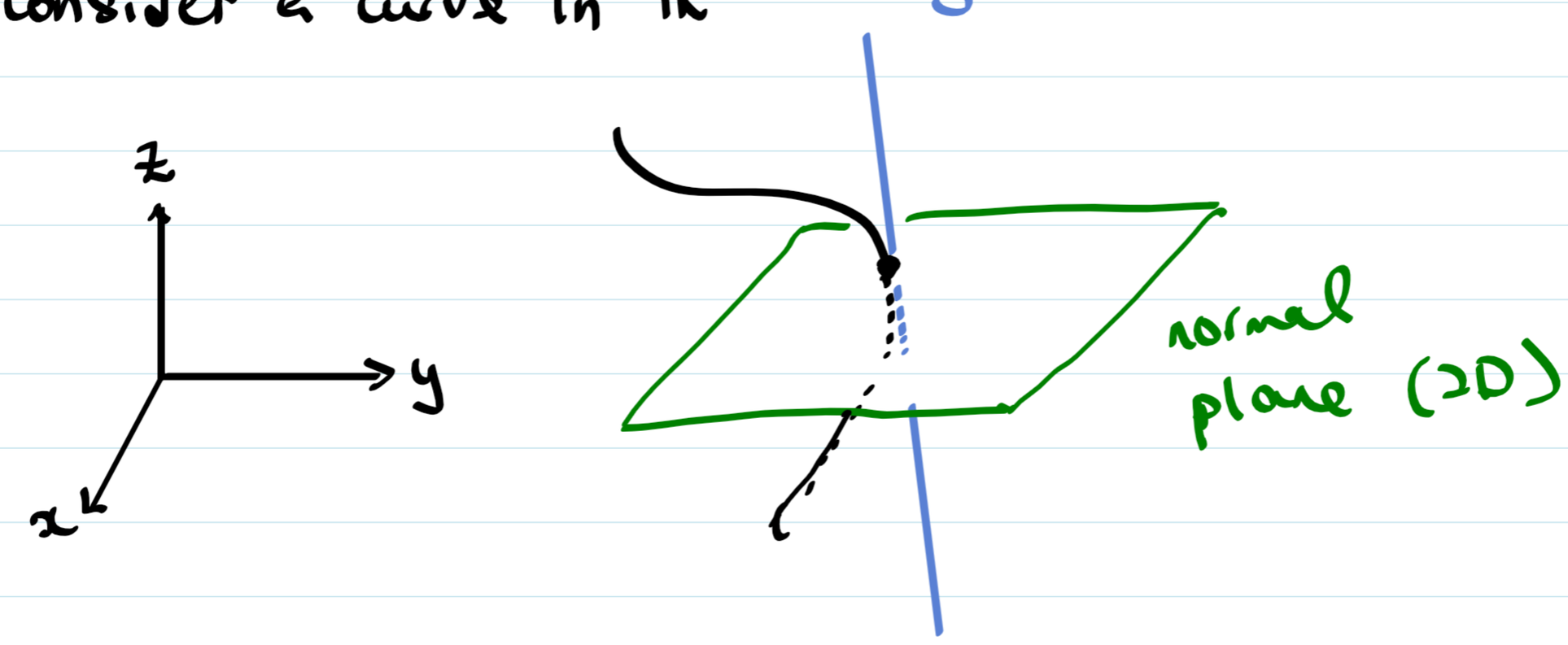
a) Curves in $\mathbb{R}^2$

Consider a curve in $\mathbb{R}^2$

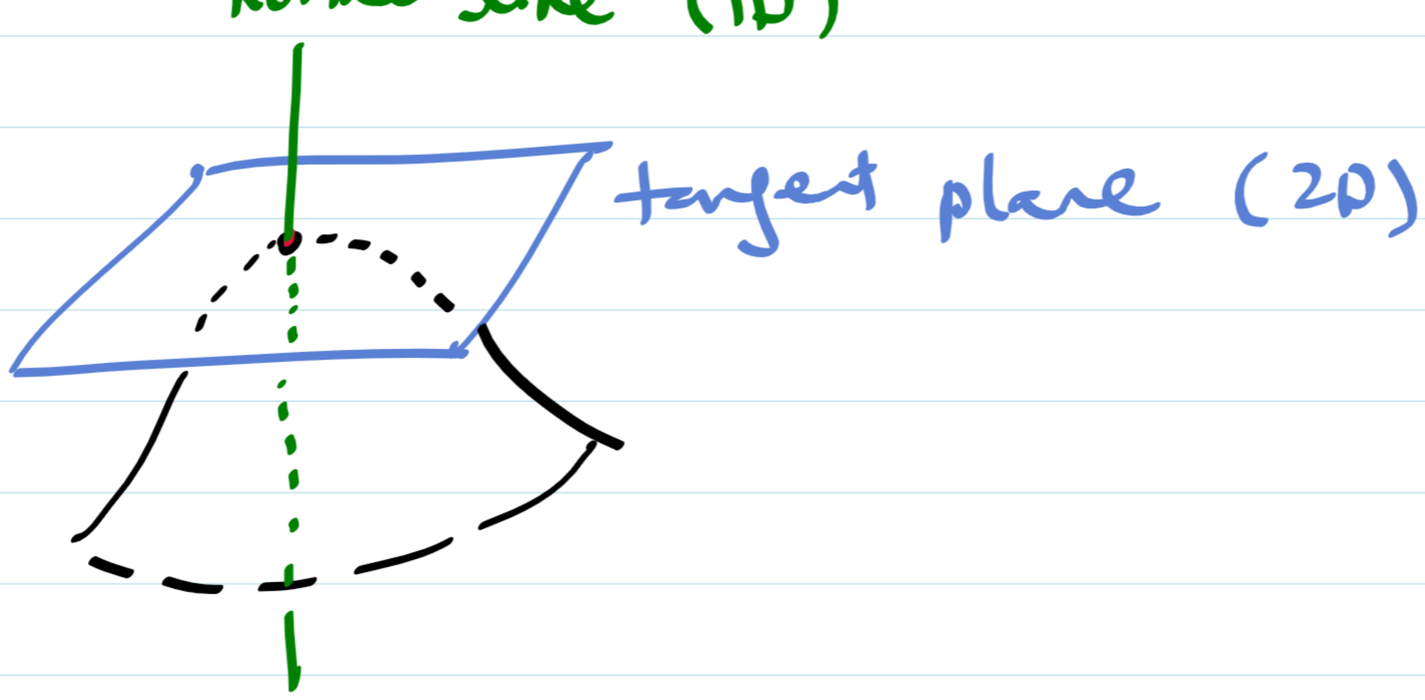


b) Curves in $\mathbb{R}^3$

Consider a curve in $\mathbb{R}^3$



c) Surfaces in $\mathbb{R}^3$ (now)



13.6.1 Tangent Planes and Normal Lines

Consider a level surface $S: F(x,y,z) = c$ ($c = \text{const}$)

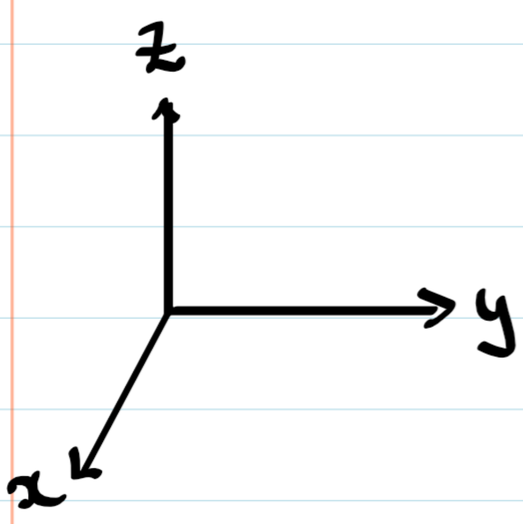
Consider a curve C on S given by

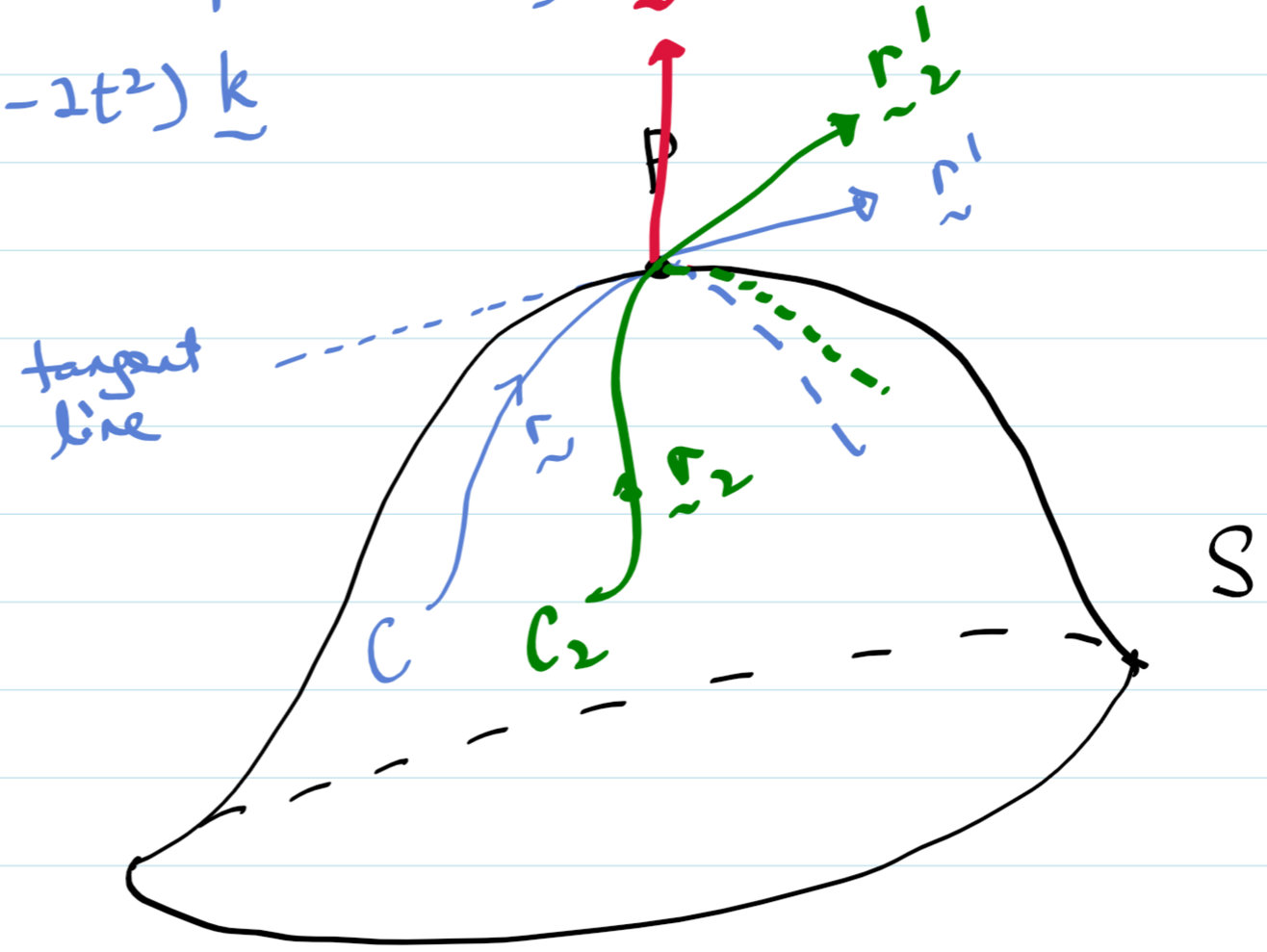
$$\underset{\sim}{r}(t) = x(t)\underset{\sim}{i} + y(t)\underset{\sim}{j} + z(t)\underset{\sim}{k}$$

Assume C passes thro point P

eg) $S: x^2 + y^2 + z = 2$ (paraboloid)

$$\underset{\sim}{r}(t) = t\underset{\sim}{i} + t\underset{\sim}{j} + (2-2t^2)\underset{\sim}{k}$$





$$F(\underset{\sim}{r}(t)) = c$$

$$F(x(t), y(t), z(t)) = c, \quad (c: \text{const})$$

$$\frac{d}{dt} F(\underset{\sim}{r}(t)) = 0$$

By Chain Rule

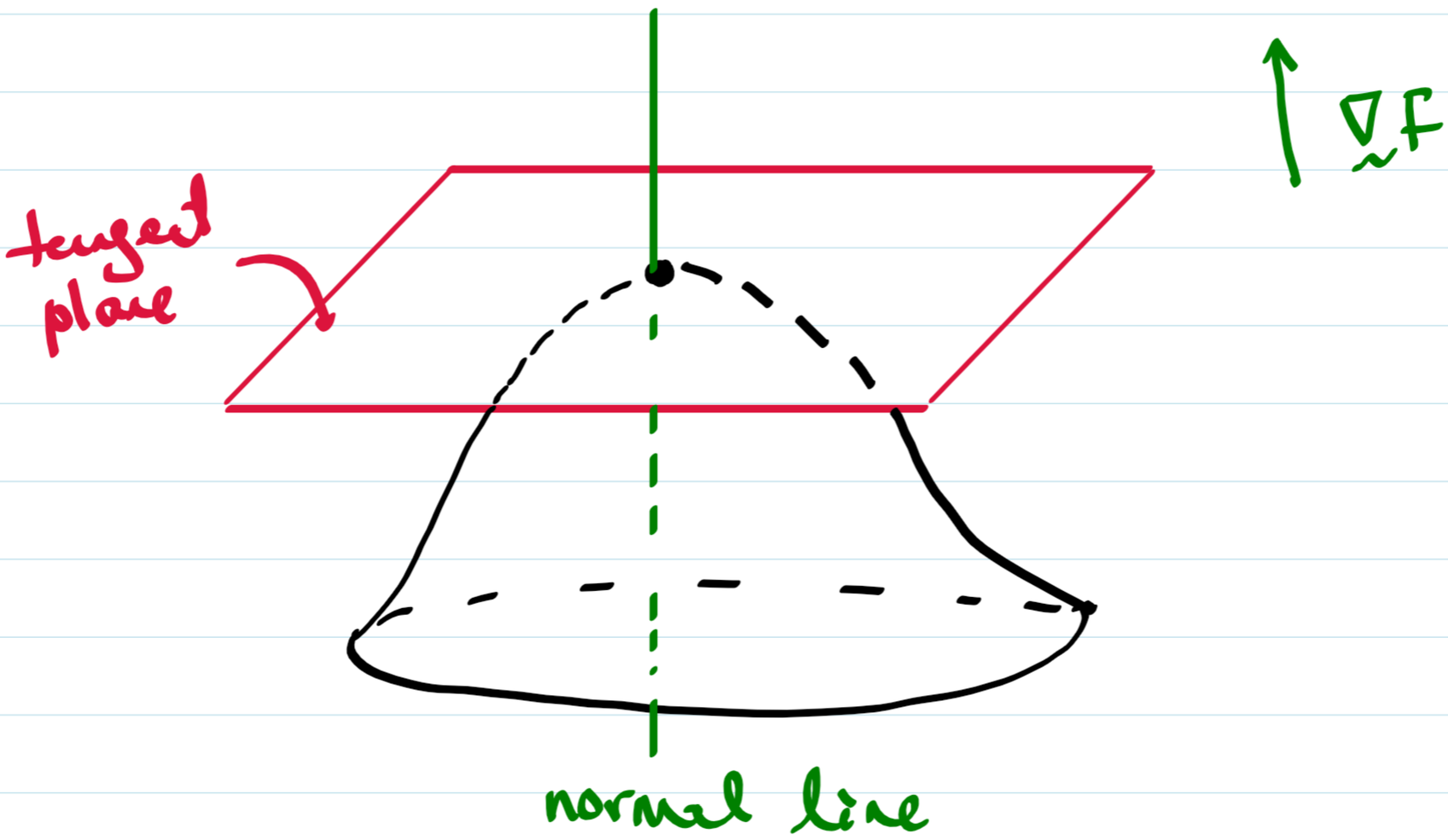
$$\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} = 0$$

$$\left(\frac{\partial F}{\partial x}\underset{\sim}{i} + \frac{\partial f}{\partial y}\underset{\sim}{j} + \frac{\partial f}{\partial z}\underset{\sim}{k}\right) \cdot \left(\frac{dx}{dt}\underset{\sim}{i} + \frac{dy}{dt}\underset{\sim}{j} + \frac{dz}{dt}\underset{\sim}{k}\right) = 0$$

$$\underset{\sim}{\nabla} f \cdot \frac{d\underset{\sim}{r}}{dt} = 0 \quad \left(\text{means } \underset{\sim}{\nabla} f \perp \frac{d\underset{\sim}{r}}{dt}\right)$$

Defn: At point P in surface S

Tangent vectors of all curves in S thro P "generate" a plane - the **tangent plane** at pt P to the surf S



Note:

a) Tangent Plane (to the surf S at point P)

- thro point P
- $\perp$ $\underset{\sim}{N} = \nabla F(P)$

b) Normal Line (to the surf S at point P)

- thro point P
- $\parallel$ $\underset{\sim}{d} = \nabla F(P)$

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(a) (b)

Example. Find equations for the tangent plane and normal line at the point P_0 on the given surface.

13.6.6. $x^2 - xy - y^2 - z = 0, \quad P_0(1, 1, -1)$.

$S: \quad f(x,y,z) = x^2 - xy - y^2 - z = 0$

Soln

Given $f(x,y,z) = x^2 - xy - y^2 - z$, $P(1\ 1\ -1)$

$\nabla f = (2x - y)\,\underset{\sim}{i} + (-x - 2y)\,\underset{\sim}{j} - \underset{\sim}{k}$

$\nabla f(P) = \underset{\sim}{i} - 3\underset{\sim}{j} - \underset{\sim}{k}$

a) Eqn of tangent plane (at pt $P(1\ 1\ -1)$ to surf S)

$0 = \nabla f(P) \cdot [\underset{\sim}{r} - \vec{OP}]$

$0 = x - 3y + z - 1 + 3 - 1, \quad 0 = x - 3y + z + 1$

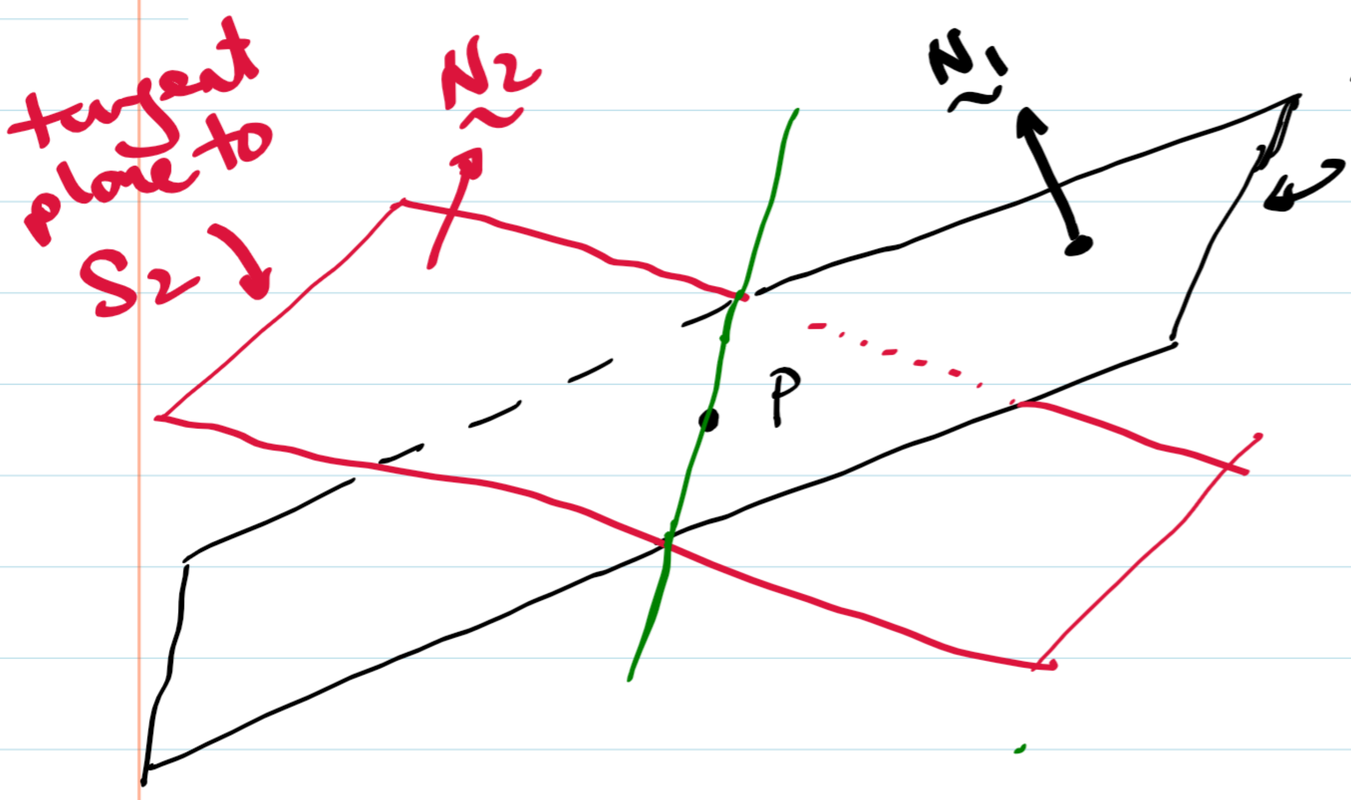
'b) Eqn of Normal line (. . . . .) :

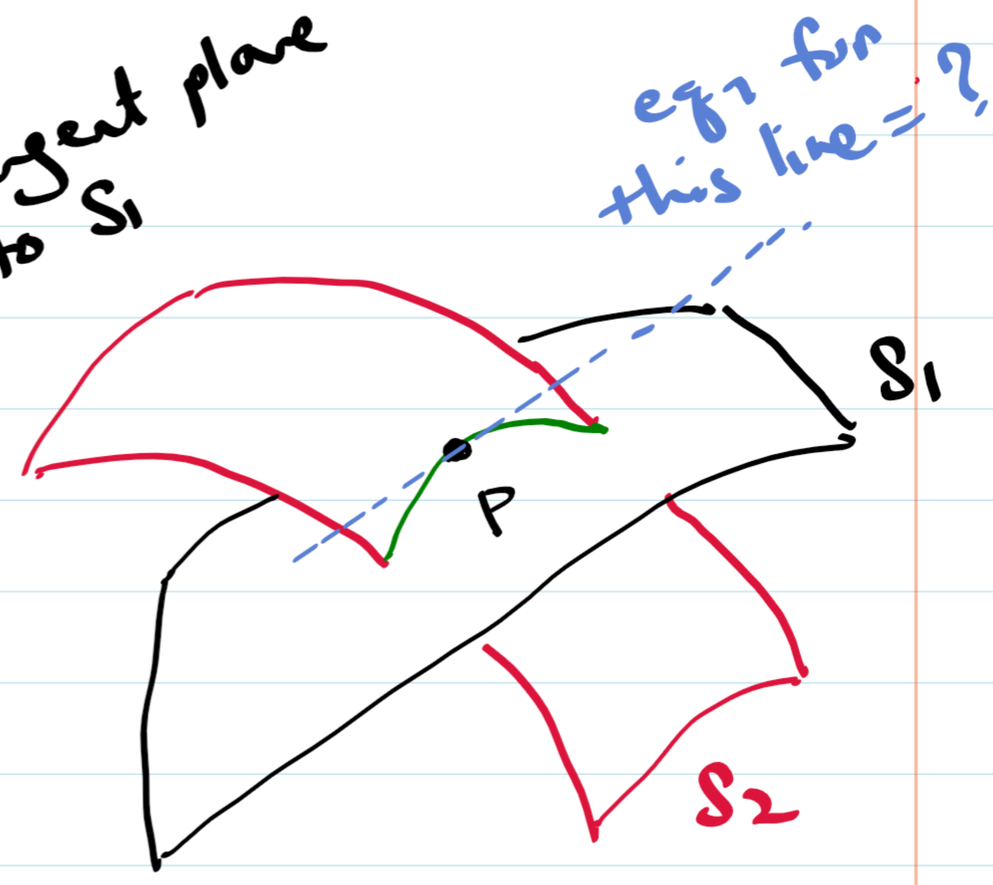
$x = 1 + t, \quad y = 1 - 3t, \quad z = -1 - t$

Example. Find parametric equations for the line tangent to the curve of intersection of the surfaces at the given point.

S_1 S_2

13.6.16. $xyz = 1, \quad x^2 + 2y^2 + 3z^2 = 6, \quad$ Point: $(1,1,1)$.





(#) the "Green line" $\parallel \underset{\sim}{N_1} \times \underset{\sim}{N_2}$

(#) We can take $\underset{\sim}{N_1} = \underset{\sim}{\nabla} F_1(P)$, $\underset{\sim}{N_2} = \nabla F_2(P)$

Soln Given Surf $S_1 : F_1(xyz) = xyz = 1$

$S_2 : F_2(xyz) = x^2 + 2y^2 + 3z^2 = 6$

at $P(1\,1\,1)$

Take $\underset{\sim}{N_1} = \underset{\sim}{\nabla} F_1(P) =$ _____

$= \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$

$\underset{\sim}{N_2} = \underset{\sim}{\nabla} F_2(P) =$ _____

$= 2\underset{\sim}{i} + 4\underset{\sim}{j} + 6\underset{\sim}{k}$

$$\underset{\sim}{d} = \underset{\sim}{N_1} \times \underset{\sim}{N_2} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 1 & 1 & 1 \\ 2 & 4 & 6 \end{vmatrix} = 2\underset{\sim}{i} - 4\underset{\sim}{j} + 2\underset{\sim}{k}$$

$\therefore$ eqn of line ..... is

$x = 1 + 2t, \quad y = 1 - 4t, \quad z = 1 + 2t.$

Example (13.6.24). By about how much will

$$h(x,y,z) = \cos(\pi xy) + xz^2$$

change if the point $P(x,y,z)$ moves from $P_0(-1,-1,-1)$ a distance of $ds = 0.1$ unit toward the origin?

a) the Ch of h, Δh = ?
when particle moving from P_0 toward origin
after a dist of 0.1



$$\Delta h \cong \frac{dh}{ds} \cdot \Delta s$$

$$\Delta h \cong D_{\underline{u}} h \cdot \Delta S$$

$\Delta S = 0.1$

Example (13.6.24). By about how much will

$$h(x,y,z) = \cos(\pi xy) + xz^2$$

change if the point $P(x,y,z)$ moves from $P_0(-1,-1,-1)$ a distance of $ds = 0.1$ unit toward the origin?

Soln

$$\nabla h = \left[-\pi y \sin(\pi xy) + z^2\right]\underset{\sim}{i} - \pi x \sin(\pi xy)\underset{\sim}{j} + 2xz\underset{\sim}{k}$$

$$\nabla h(P_0) = \underset{\sim}{i} + 2\underset{\sim}{k}$$

Take $\underset{\sim}{u} = \overrightarrow{P_0O} = \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$

$$\hat{\underset{\sim}{u}} = \frac{\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}}{\sqrt{3}}$$

$\therefore$ the small ch in h

$$= D_{\hat{\underset{\sim}{u}}} h(P_0) \cdot \Delta S = \frac{3}{\sqrt{3}} \cdot (0.1) \simeq +0.17$$

$\therefore$ h will increases by 0.17 unit when P moves ..... towards the origin.